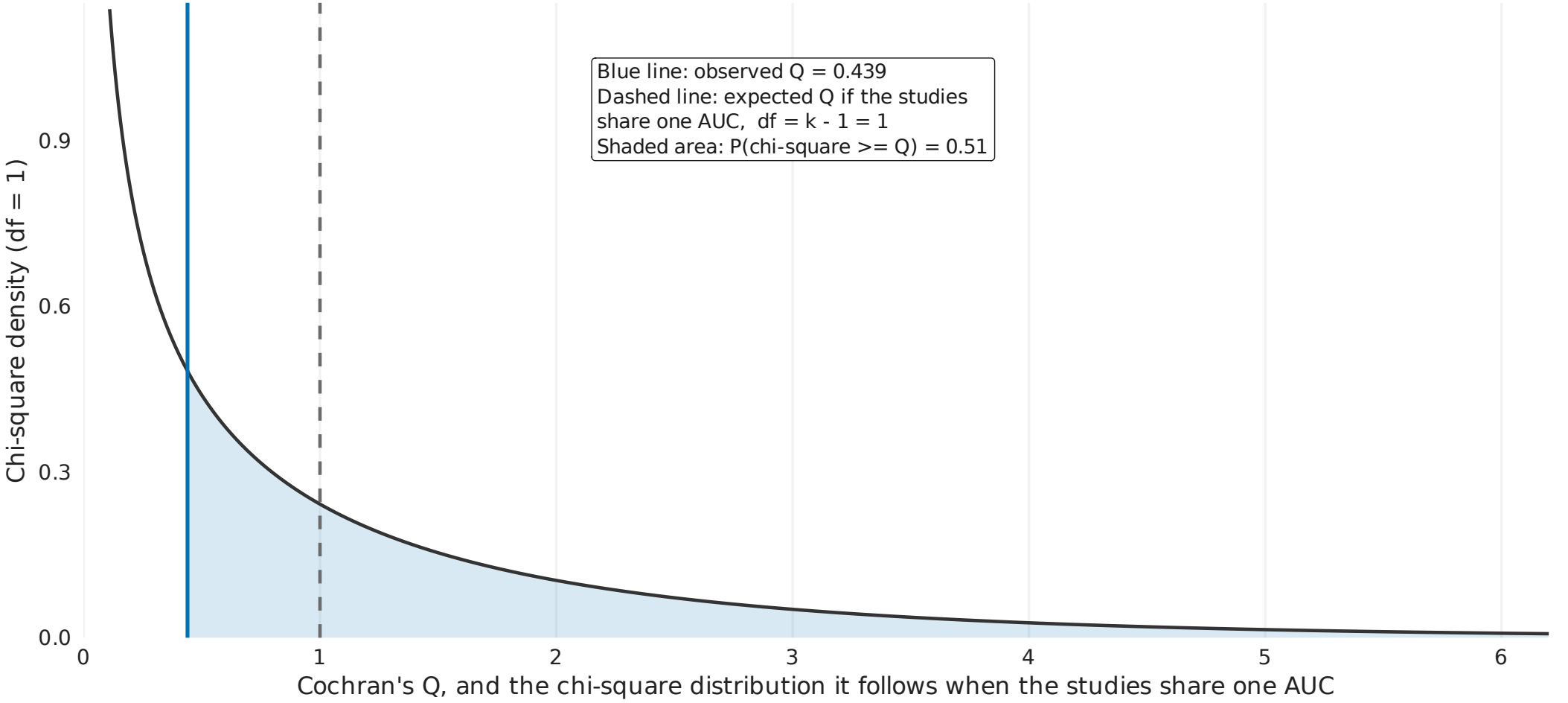
5. Under pure sampling error, Q averages df, not zero

Two studies give one contrast, so df = 1. Chance alone produces a Q whose average is 1, not 0.
The median of this curve is 0.45. The observed Q, 0.44, sits on that median.



The heterogeneity p-value is the shaded area: how often chance alone would produce a Q at least this large. Here, about half the time.
The density rises without bound as Q approaches 0. The panel is cut at 1.15 so the rest of the curve stays visible.